

ence, reference designs are available. App notes on TI's centre address ways to lower power consumption, through software examples for all profiles in the BLE stack. TI's unique 'Over the Air Download' enables future firmware upgrades wirelessly in the field."

Navaneethan shares that there are three main things that ease the process of designing with BLE: "One would be to use pre-verified, pre-certified Bluetooth modules that integrate antenna, power supplies and MCUs with the complete software stack. These modules provide a drop-in functionality that can be added to any embedded board at the PCB utilising UART, PC or GPIO interfaces."

"Readily available consumer devices like the Samsung Galaxy S4 and the Apple iPhone 4S and 5 also include Bluetooth stacks and thus can be used for testing embedded designs. In general, Android devices are preferred due to their greater flexibility," adds Navaneethan.

Reducing interference

As mentioned earlier, Bluetooth Smart operates in the 2.4-2.485GHz band using a spread-spectrum, frequency-hopping and full-duplex signal at a nominal rate of 1600 hops/second.

Since there would be multiple wireless technologies using the mostly unlicensed 2.4GHz spectrums, interference here can be reduced by the adaptive frequency hopping capability that is designed into the technology. It works by detecting the other devices in the spectrum as well as the frequencies

Development kits for BLE

Texas Instruments has come out with the CC2540 development kit, which includes CC2540-based modules and dongles, and general-purpose development boards for hardware prototyping, along with the necessary antennae, cables and documents. This is a quick way to get familiar with designing using the CC2540 SoCs mentioned earlier in the article.

EM Microelectronics has also made available the ENSEMBLE BLE development kit, which allows designers and developers to test their own BLE application on a small, easy-to-use, portable hardware platform.

The ALPWF-BLESDBK80c51 software development kit from Alpwise provides a compatible BLE stack for the EM9301, as well as many profiles, services and example applications for the ENSEMBLE DVK.

The nRFgo starter kit and the nRF8001 development kit from Nordic Semiconductor is one way to jumpstart development using the nRF8001 BLE IC. The kit also includes a BLE emulator board, which connects to a PC via USB to enable monitoring and control of peer-to-peer connections.

From CSR, the μ Energy development kit includes a complete software development environment with graphical code development and debug facilities. The SDK includes extensive example applications and application notes together with a qualified Bluetooth 4.0 stack.

that they are using. Once the frequencies are detected, the device in question can avoid those frequencies in its operation. The best part is that it works with even other wireless technologies apart from Bluetooth.

Ensuring that your product succeeds

Ensuring market success is as important as designing it. Navaneethan suggests, "When designing a new product with Bluetooth for the first time, focus on getting to market faster than on attaining the lowest cost possible. Reduce development/go-to-market time by starting with complete Bluetooth development kits from chip or module vendors that get closer to your eventual design needs—even if they are more expensive. You can reduce

the cost in the next iteration of your product release."

As an example, if you intend to support iOS or Android devices or apurricular application like audio over Bluetooth, choose a module or a kit that already has proven design examples with these interfaces.

"Increase reliability by using pre-verified and pre-certified Bluetooth modules. Do not underestimate the time/effort it would take to do wireless certification for your own designs. If you are designing Bluetooth 4.0 system along with USB 3.0 on the same board, place these components as far away as possible" adds Navaneethan.

The most critical aspect of any wireless design is good PCB design/layout and assembly configuration of rest of the peripheral modules. Any small mistake in doing that can lead to non-working of your design.

"The worst thing that could happen in such a case is that the project works sometimes, and sometimes it doesn't. This is a reliability issue and it is extremely difficult to detect the pattern and root cause. Taking precautions during integration itself would save a lot of resources. In a nutshell, read the user manuals and component datasheets carefully and thoroughly before getting hands on," adds Anand. ●

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Mobile OS support for BLE

If you are designing a device for use with smartphones, then this section is for you. Apple devices have supported BLE from the fifth iteration of their iOS operating systems iOS 5, and on their OS X as well. Blackberry 10 also features support for BLE.

Android, on the other hand, has been slightly slower. It has just come out and announced support for BLE in May 2013. The announcement confirms that BLE will be supported in Android API Level 18 and higher versions.

It is interesting to note that while Google was late in implementing BLE in Android, Samsung and HTC developed their own SDKs for BLE so that they are not left behind in what is an obvious evolution in the use of consumer electronics.

Finally, Windows Phone 8 supports only Bluetooth 3.1. However, the chipset (WCN3660) used in Lumia 820 and 920 phones does support BLE. Thus it is expected to receive a stack update that will enable it to feature in those devices and thus herald Windows Phone 8 to the world of BLE. Microsoft does support BLE on its desktop operating system.